

PFMEA Training Guide - AIAG-VDA 2019

Complete Learning Path | Quality Head Agents

Module 1: FMEA Fundamentals (2 hours)

What is FMEA and why it matters. History: from NASA/military to automotive (AIAG, VDA). Five Core Tools and FMEA role. Types: DFMEA, PFMEA, DRBFM. IATF 16949 requirements. Key terminology: Failure Mode, Effect, Cause, Severity, Occurrence, Detection, Action Priority.

Module 2: PFMEA Structure - AIAG-VDA 2019 (3 hours)

Step 1: Project Identification. Step 2: Structure Analysis. Step 3: Function Analysis. Step 4: Failure Analysis. Step 5: Risk Analysis (S, O, D, AP). Step 6: Optimization. Step 7: Results Documentation. Each step includes practical exercises.

Module 3: Rating Tables in Depth (2 hours)

Severity table 1-10 with automotive examples. Occurrence table 1-5. Detection table 1-5. Action Priority matrix (H/M/L). Common rating mistakes and calibration exercises. Team consensus rating techniques.

Module 4: Special Characteristics (2 hours)

CC (Critical) vs SC (Significant Characteristics). Customer-specific definitions (Ford, GM, Stellantis). Drawing symbol requirements. PFMEA and Control Plan requirements. IATF audit expectations for special characteristics management.

Module 5: Practical Workshop - Live PFMEA (3 hours)

Team exercises using real process. Build PFMEA from blank PFD. Rate all failure modes as a team. Identify AP=H items and create action plans. Link to Control Plan. Simulated auditor questions and responses.

Certification Exam

50 questions covering all 5 modules. Minimum pass: 70%. Certificate valid 3 years. Recommended for: QA Engineers, Quality Managers, Manufacturing Engineers, FMEA Team Leaders.